Deep learning [3]

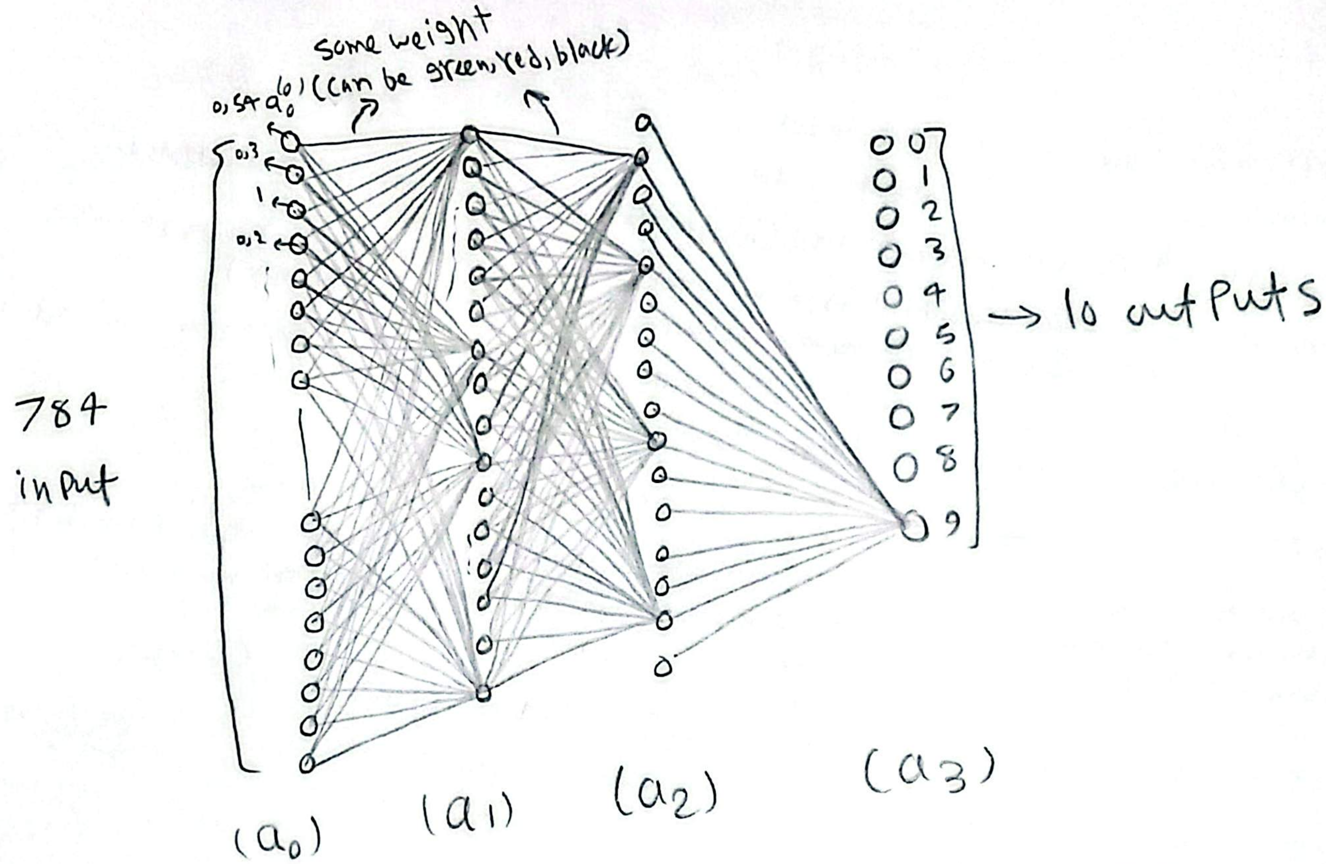
application of the neural network

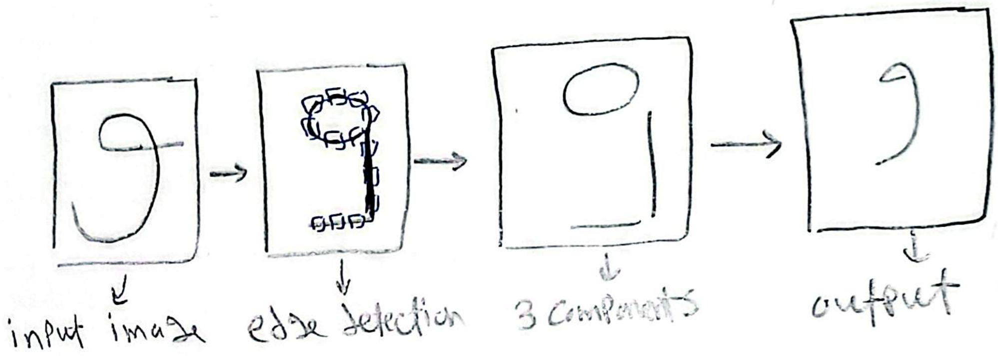
Quick recap:

$$a^{(n)} = \sigma\left(W a^{(n-1)} + b\right)$$ → easy to apply in coding

1 neuron: a function that takes the outputs of all neurons in the previous layer a[nd] gives a number between $0 \rightarrow 1$.

network: a function that takes 784 number as an input and outputs 10 possible
" a complex function that contains 13,000 parameters".

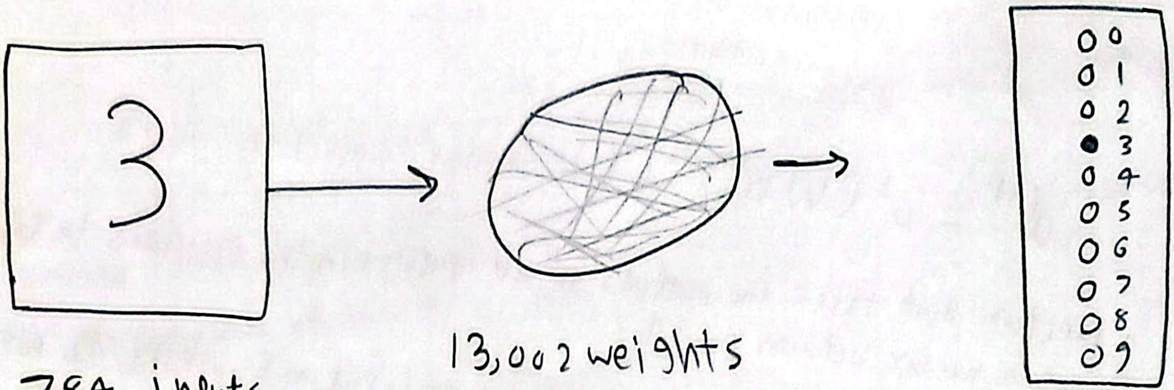




* in modern neural networks, sigmoid function is barely used, they use ReLU as it's much easier to train. ReLU → rectified Linear unit.

$$\mathrm{ReLU}(a) = \max(0, a)$$

how we train a neural networks ?



784 inputs — 13,002 weights and biases — 10 outputs

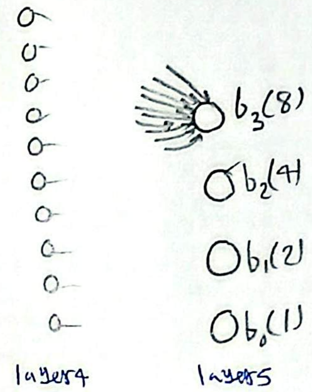
Terminology

1. Left most layer is the input layer
2. right ———— output layer
3. middle layers are hidden layers (hidden because they are not inputs or outputs).
4. neural networks where the output from one layer is used as input to the next layer this is called feedforward neural networks (no loops in this network).
5. if there's a loop, so the loop make the input depends on the output, called feedback network also called recurrent neural networks.

exercise

if we want to add a new layer after the output layer, which outputs the bitwise representation of the digit so the new layer have 4 neurons only, what are the weights and bias for this new layer, assume perfect trained old output layer.

Solution:



perfect layer 4 : means if the result is 4 so [0 0 0 0 1 0 0 0 0 0]

weights : we gonna make a fixed weights matrix for each digit :

[0000, 0001, 0010, 0011, ————]

bias = −0,5

so if the result from layer 4 is [0 0 0 0 1 0 0 0 0 0]

$b_3 = 0 \times 1 - 0{,}5 = \overset{-ve}{-0{,}5} \rightarrow \sigma \rightarrow \overset{<0{,}5}{0{,}38} \rightarrow 0$

$b_2 = 1 \times 1 - 0{,}5 = \overset{+ve}{0{,}5} \rightarrow \sigma \rightarrow \overset{0{,}5<}{0{,}622} \rightarrow 1$

$b_1 = 0 \times 1 - 0{,}5 = -0{,}5 \rightarrow \sigma \rightarrow 0{,}38 \rightarrow 0$

$b_0 = 0 \times 1 - 0{,}5 = -0{,}5 \rightarrow \sigma \rightarrow 0{,}38 \rightarrow 0$

result = 0100

Same logic with other digits.

note : in perceptrons, active condition was $w \cdot x + b > 0$, in sigmoid, active condition is $\sigma(w \cdot x + b) > 0{,}5$